

Lecture 13

Revision 02 - Spectrum

MH1201 Linear Algebra 2

What does the spectrum of a matrix refer to?

Google the answer. 😊

1. The set of determinants
- ~~2.~~ The set of eigenvalues

QUESTION 2.

(20 marks)

Let $a \in \mathbb{R}$. Consider the following 3×3 -matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & a & 1 \\ 0 & 0 & 2 \end{bmatrix}.$$

- (a) Write down the characteristic polynomial for $\mathbf{A}$ and determine its eigenvalues.
- (b) Find all values of a for which $\mathbf{A}$ is *not* diagonalizable.
- (c) For all a , for which $\mathbf{A}$ is diagonalizable, find a diagonal matrix $\mathbf{D}$ and an invertible matrix $\mathbf{P}$ such that $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$. In other words, diagonalize $\mathbf{A}$.

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & a & 1 \\ 0 & 0 & 2 \end{bmatrix}.$$

(a) Write down the characteristic polynomial for A and determine its eigenvalues.

$$\det(A - \lambda I) = (1 - \lambda)(a - \lambda)(2 - \lambda)$$

eigenvalues are $\lambda = 1, 2, a$

$$\text{Spectrum} = \{1, 2, a\}$$

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & a & 1 \\ 0 & 0 & 2 \end{bmatrix}.$$

Which of the following statement is TRUE?

1. The geometric multiplicities of A are 1, 2, a .
2. The algebraic multiplicities of A are 1, 2, a .
- ~~3.~~ Statements 1&2 are not true.

① Matrix has eigenvalues 1, 2, a .

Each eigenvalue has geom multiplicity + alg. multiplicity.

$\dim(E_\lambda) = \text{positive integer}$

count no. of times λ appears as a root in char. poly.

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & a & 1 \\ 0 & 0 & 2 \end{bmatrix} .$$

char poly
 $(1-\lambda)(a-\lambda)(2-\lambda)$

Which of the following statement is TRUE?

1. The algebraic multiplicity of 1 is ALWAYS one. i.e. $\alpha(1) = 1$.
2. The algebraic multiplicity of 2 is ALWAYS one. i.e. $\alpha(2) = 1$.
3. The algebraic multiplicity of a is ALWAYS one. i.e. $\alpha(a) = 1$.
- ~~4.~~ Statements 1&2&3 are not true.

we don't know what a is.

$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & a & 1 \\ 0 & 0 & 2 \end{bmatrix}$. (b) Find all values of a for which A is *not* diagonalizable.

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$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & a & 1 \\ 0 & 0 & 2 \end{bmatrix} .$$

(c) For all a , for which $\mathbf{A}$ is diagonalizable, find a diagonal matrix $\mathbf{D}$ and an invertible matrix $\mathbf{P}$ such that $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$. In other words, diagonalize $\mathbf{A}$.

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QUESTION 4.

(20 marks)

Let J be the $(n \times n)$ -matrix that comprises all ones.

- (a) Let $\mathbf{j} = (1, 1, \dots, 1)^T$. Explain why $\mathbf{j}$ is an eigenvector of J . In your explanation, determine the corresponding eigenvalue.
- (b) Determine a basis for $\text{null}(J)$. ~~Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be defined by $T(\mathbf{x}) = J\mathbf{x}$.~~
 - ~~(i) Explain why T is a linear operator.~~
 - ~~(ii) Determine a basis for $\text{null}(T)$.~~
- (c) Diagonalize J .
- (d) Consider the following $(n \times n)$ -matrix

$$A = \begin{bmatrix} 0 & 1 & \cdots & 1 \\ 1 & 0 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 0 \end{bmatrix}.$$

In other words, $A_{ij} = 0$ if $i = j$ and $A_{ij} = 1$, otherwise. Diagonalize A .

Let $\mathbf{J}$ be the $(n \times n)$ -matrix that comprises all ones.

- (a) Let $\mathbf{j} = (1, 1, \dots, 1)^T$. Explain why $\mathbf{j}$ is an eigenvector of $\mathbf{J}$. In your explanation, determine the corresponding eigenvalue.
-

- (b) Determine a basis for $\text{null}(\mathbf{J})$. Let $\mathbf{T} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be defined by $\mathbf{T}(\mathbf{x}) \equiv \mathbf{J}\mathbf{x}$.
- (i) Explain why $\mathbf{T}$ is a linear operator.
 - (ii) Determine a basis for $\text{null}(\mathbf{T})$.
-

(b) Determine a basis for $\text{null}(\mathbf{J})$. Let $\mathbf{T} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be defined by $\mathbf{T}(\mathbf{x}) = \mathbf{J}\mathbf{x}$.

(i) Explain why $\mathbf{T}$ is a linear operator.

(ii) Determine a basis for $\text{null}(\mathbf{T})$.

(c) Diagonalize $\mathbf{J}$.

(c) Diagonalize J .

(d) Consider the following $(n \times n)$ -matrix

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & \cdots & 1 \\ 1 & 0 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 0 \end{bmatrix}.$$

In other words, $A_{ij} = 0$ if $i = j$ and $A_{ij} = 1$, otherwise. Diagonalize $\mathbf{A}$.

Set $n = 3$. Then

$$\begin{aligned} \mathbf{A} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} &= \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix}^{-1} \\ &= \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \left(\frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ -2 & 1 & 1 \\ 1 & -2 & 1 \end{bmatrix} \right) \end{aligned}$$

Set $n = 3$. Then

HIDDEN

When $k = 20$, what is A^{20} approximately?

More generally...

$$\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$$

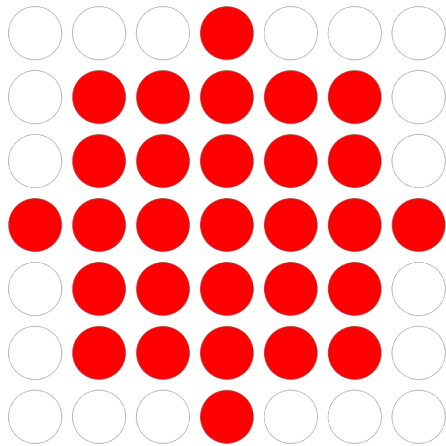
$$= \begin{bmatrix} \uparrow & & \uparrow \\ \mathbf{c}_1 & \cdots & \mathbf{c}_n \\ \downarrow & & \downarrow \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_n \end{bmatrix} \begin{bmatrix} \leftarrow & \mathbf{r}_1 & \rightarrow \\ & \vdots & \\ \leftarrow & \mathbf{r}_n & \rightarrow \end{bmatrix}$$

$$= \lambda_1 \mathbf{M}_1 + \lambda_2 \mathbf{M}_2 + \cdots + \lambda_n \mathbf{M}_n$$

Consider the following 7×7 -matrix A .

Eigenvalues of A : $5.08, -1.30, 1.21, 1.77e-16, -7.89e-18, 0.73e-33, -2.33e-34$

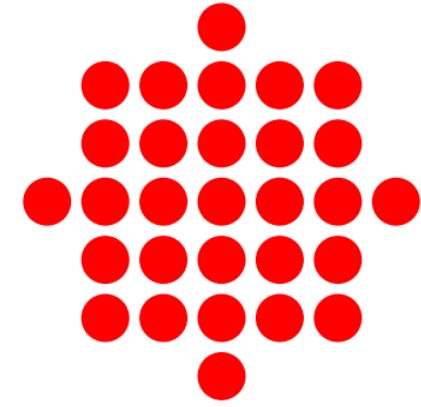
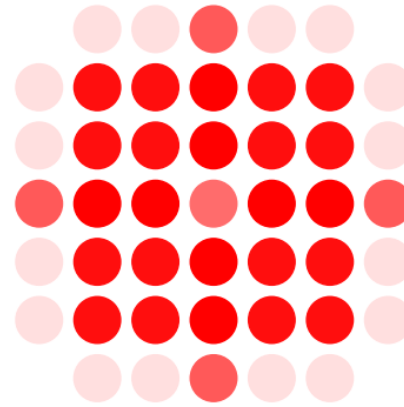
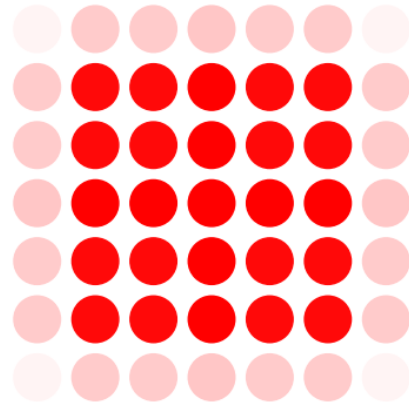
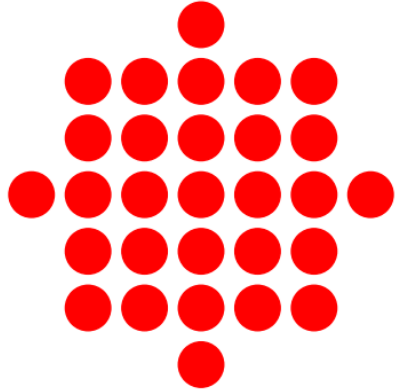
And A is diagonalizable.



Consider the following 7×7 -matrix A .

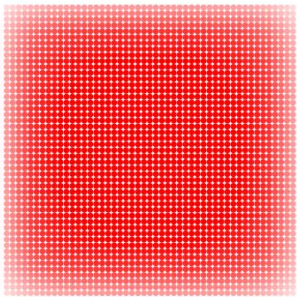
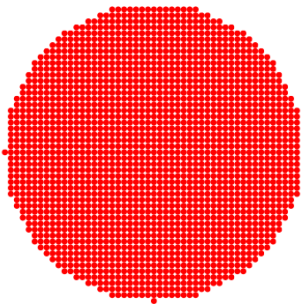
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And A is diagonalizable. i.e.



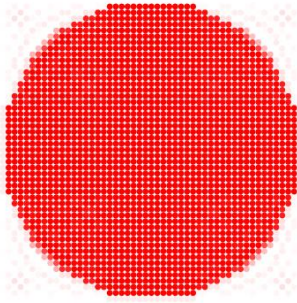
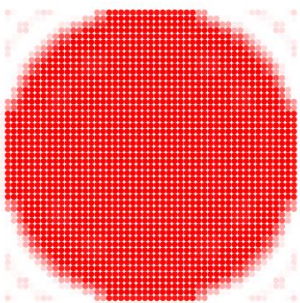
Consider the following 50×50 -matrices...

LGC; PI



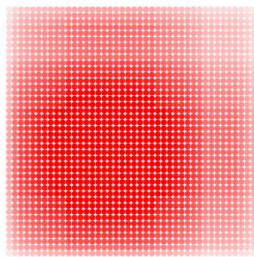
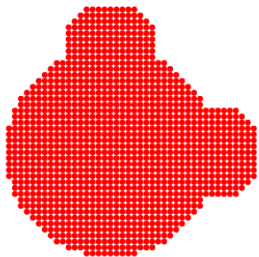
Original Matrix

Top 1 eigenvalue.



Top 5 eigenvalues.

Top 10 eigenvalues.



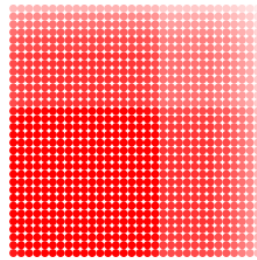
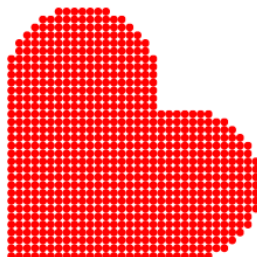
Original Matrix

Top 1 eigenvalue.



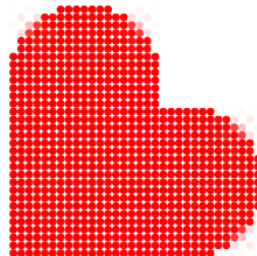
Top 5 eigenvalues.

Top 10 eigenvalues.



Original Matrix

Top 1 eigenvalue.



Top 5 eigenvalues.

Top 10 eigenvalues.

Things not explained...

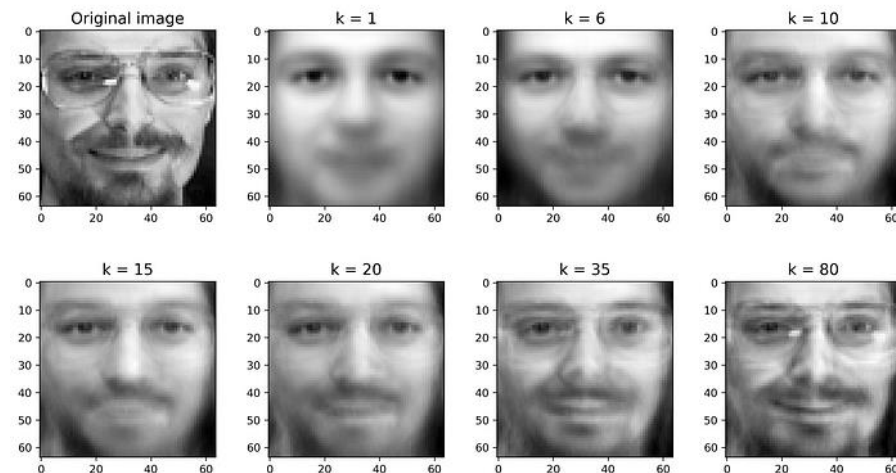
TA; PI

- How to handle non-square matrices?
- Why is it more efficient to store the M_i 's instead of A ?
- How to find larger eigenvalues quickly?

Other resources

- Very instructive article by Reza Bagheri (<https://towardsdatascience.com/understanding-singular-value-decomposition-and-its-application-in-data-science-388a54be95d>)
- Gilbert Strang's Linear Algebra course (<https://math.mit.edu/~gs/linearalgebra/>)

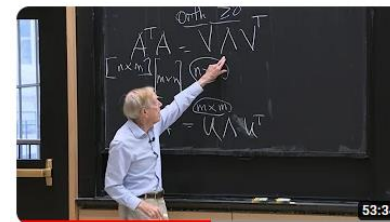
Reconstructed image using the first k singular values



Gilbert Strang lectures on Linear Algebra (MIT)
Robert Talbert
2. Elimination with Matrices • 47:42
3. Multiplication and Inverse Matrices • 46:49
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MIT 18.06SC Linear Algebra, Fall 2011
MIT OpenCourseWare
An Interview with Gilbert Strang on Teaching Linear Algebra • 7:34
Course Introduction | MIT 18.06SC Linear Algebra • 7:13
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6. Singular Value Decomposition (SVD)
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